

Block G Certificate

Lean Verification

VERDICT: PASS_WITH_OBSERVATIONS

Scope

Verify the Lean 4 formalization compiles cleanly and that the axiom footprints of each node match the manuscript's claims.

Method

Built the frozen Lean project (8060 compilation jobs) and extracted per-node axiom footprints via `#print axioms`.

Findings

- **8060 jobs** compiled with **0 errors**.
- Exactly **2 axioms** in use: AX1 and AX2.
- Corridor nodes verified **closed** (standard axioms only).
- **1 minor observation (F-G05)**: sparse node E_8 carries AX1 + AX2, not AX2-only as §11.6 states (it reuses bulk lemma E_4_3 on $\alpha \in [1/12, 1/2]$). Attribution imprecision only, no mathematical impact.

Evidence pointers

RESULTS/blockG_lean_verification/README.md

8060 jobs, 0 errors, axioms {AX1, AX2}

Observation F-G05 (minor, non-blocking)

Lean freeze ZIP SHA-256: 060957e6b8d54779844dc6adf7cc7c3b8446fc17a87aa8d7a437e9d9d1001b78

Attestation

Auditor	Claude Opus 4.8 (Anthropic)
Timestamp	2026-07-28
Manuscript SHA-256	7aaf03083ddf7731dcb2b1e849cdfac97fb1697df1650c49a56e8431ce1bcb0b
Block G hash source	README.md
Block G SHA-256	d403fd318fb1aeddfe9f7b8adbb7b983f07404fc34389563b1aaa1a409c149b0